

Day 3: Tasks

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.



BIOS / POST Error Message	Hardware or Configuration Issue
No Bootable Device Found	Hard drive/SSD not detected or operating system missing.
CMOS Checksum Error	Weak CMOS battery or corrupted BIOS settings.
Keyboard Error / No Keyboard Present	Keyboard disconnected, faulty, or port issue.
Memory Test Failed	RAM is loose, faulty, or incompatible.
CPU Fan Error	CPU fan not connected or not spinning properly.
Disk Boot Failure	Incorrect boot order or damaged boot drive.
SMART Status Bad, Backup and Replace	Hard drive is failing and should be replaced soon.
CMOS Battery Low	CMOS battery is running out of power.
Operating System Not Found	OS is missing, corrupted, or the wrong drive is selected for booting.
USB Device Over Current Status Detected	Faulty USB device, damaged USB port, or motherboard issue.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.



POST Beep Code Scenario (AMI BIOS)

Problem:

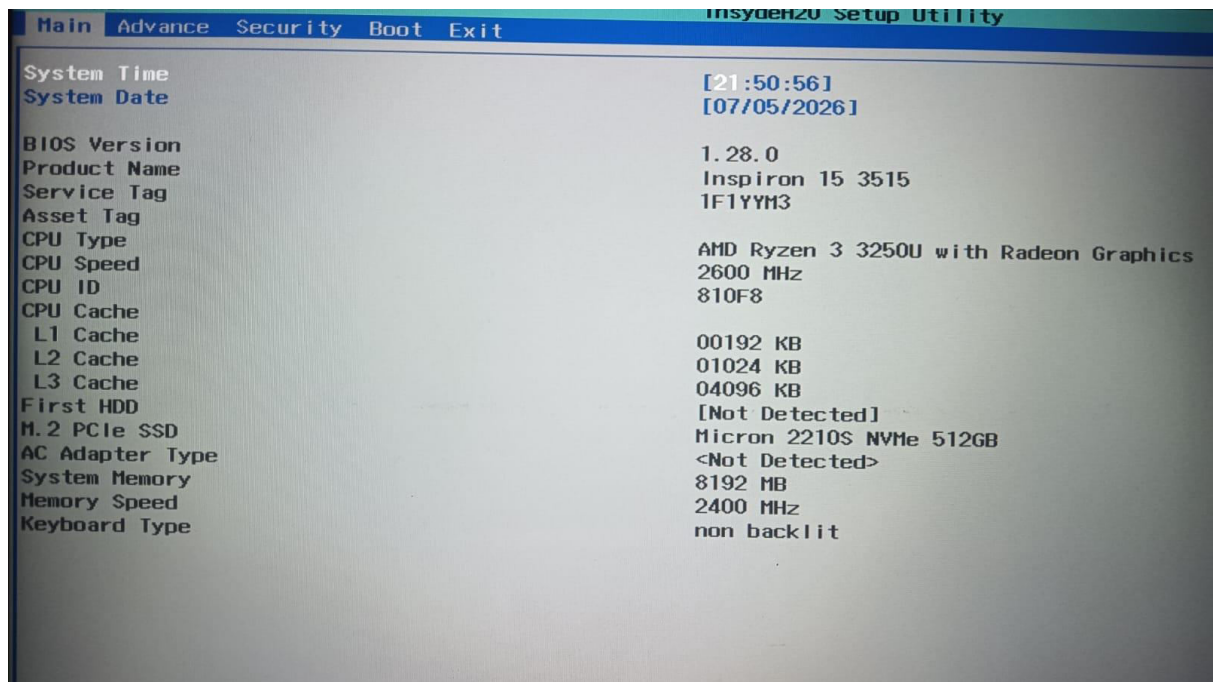
A PC beeps 3 times in a repeating pattern during startup and nothing appears on the monitor.

Typical Meaning :

3 short beeps usually indicate a **RAM failure**. The BIOS cannot properly detect or access the system memory during the Power-On Self-Test.

Troubleshooting Steps

1. **Reseat the RAM**
 - Turn off the PC and unplug the power cable.
 - Remove the RAM modules and reinstall them firmly in their slots.
 - Ensure they click into place properly.
2. **Test RAM Modules Individually**
 - If multiple RAM sticks are installed, remove all but one and try booting.
 - Test each RAM stick and slot separately to identify a faulty RAM module or motherboard slot.
3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.



CMOS Checksum Error – Defaults Loaded

Meaning:

The message "**CMOS checksum error – Defaults loaded**" indicates that the BIOS detected incorrect or corrupted data in the CMOS memory. As a result, the system loaded the default BIOS settings. This commonly happens when the **CMOS battery is weak or dead**, causing BIOS settings to be lost when the computer is powered off.

How to Resolve the Issue

➤ Enter BIOS Setup

- Restart the computer and press **Del**, **F2**, or the required key to enter BIOS.

- **Check and Configure Settings**
 - Verify the **date and time**.
 - Reconfigure any custom BIOS settings such as boot order if needed.
- **Save and Exit**
 - Save the settings and restart the computer.
- **Replace the CMOS Battery**
 - If the error returns, replace the motherboard's **CR2032 CMOS battery**.
 - After replacement, enter BIOS again and set the correct date, time, and other settings.